

Yuzheng Hu

Research Scientist, Google Research NYC

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EDUCATION

University of Illinois at Urbana-Champaign

Ph.D. in Computer Science

Urbana, IL, USA

Aug. 2021 – May. 2026

- **Advisor:** Han Zhao
- **Thesis:** Advancing Generative AI via the Science of Data

Peking University

B.S. in Mathematics (Math and Applied Math, Honor Track)

Beijing, China

Sep. 2017 – Jul. 2021

- **Advisors:** Wei Lin and Liwei Wang

RESEARCH INTERESTS

I develop the **science of data**—a perspective that views data as an *optimizable*, *expandable* and *predictable* component of AI systems. To this end, I work on **data attribution**, **synthetic data**, and **privacy**.

PUBLICATIONS

★ stands for equal contribution or alphabetical order

- [P1] MAPLE: Metadata Augmented Private Language Evolution 📄
Eli Chien*, **Yuzheng Hu***, Ryan McKenna, Shanshan Wu, Zheng Xu, Peter Kairouz
The 3rd Conference on Language Modeling (COLM 2026)
- [P2] ACTG-ARL: Differentially Private Conditional Text Generation with RL-Boosted Control 📄 🗣️
Yuzheng Hu, Ryan McKenna, Da Yu, Shanshan Wu, Han Zhao, Zheng Xu, Peter Kairouz
The 43rd International Conference on Machine Learning (ICML 2026)
- [P3] OPUS: Towards Efficient and Principled Data Selection in Large Language Model Pre-training in Every Iteration 📄 😊
Shaobo Wang, Xuan Ouyang, Tianyi Xu, **Yuzheng Hu**, Jialin Liu, Guo Chen, Tianyu Zhang, Junhao Zheng, Kexin Yang, Xingzhang Ren, Dayiheng Liu, Linfeng Zhang
The 43rd International Conference on Machine Learning (ICML 2026, Oral)
- [P4] A benchmark of expert-level academic questions to assess AI capabilities 📄 🌐 📚
Dataset contributor of Humanity’s Last Exam; authored 4 of 2,500 expert-level questions, with one selected as a prize-winning entry (**top 550 worldwide**)
Nature (2026)
- [P5] A Snapshot of Influence: A Local Data Attribution Framework for Online Reinforcement Learning 📄 🗣️
Yuzheng Hu*, Fan Wu*, Haotian Ye, David Forsyth, James Zou, Nan Jiang, Jiaqi W. Ma, Han Zhao
The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025, Oral)
- [P6] Taming Hyperparameter Sensitivity in Data Attribution: Practical Selection Without Costly Retraining 📄
Weiyi Wang, Junwei Deng, **Yuzheng Hu**, Shiyuan Zhang, Xirui Jiang, Runting Zhang, Han Zhao, Jiaqi W. Ma
The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025)
- [P7] MergeBench: A Benchmark for Merging Domain-Specialized LLMs 📄 🗣️
Yifei He, Siqi Zeng, **Yuzheng Hu**, Rui Yang, Tong Zhang, Han Zhao
The 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025, Datasets and Benchmarks Track)

- [P8] Empirical Privacy Variance  
Yuzheng Hu*, Fan Wu*, Ruicheng Xian, Yuhang Liu, Lydia Zakynthinou, Pritish Kamath, Chiyuan Zhang, David Forsyth
The 42nd International Conference on Machine Learning (ICML 2025)
- [P9] An Improved Autoregressive Evaluation Paradigm for Large Language Models 
Jipeng Zhang*, Rui Pan*, **Yuzheng Hu***, Kashun Shum*, Guanyu Yao, Xiang Liu, Renjie Pi, Hanze Dong, Shizhe Diao, Yong Lin, Han Zhao, Tong Zhang
ACM Transactions on Intelligent Systems and Technology (ACM TIST 2025)
- [P10] Most Influential Subset Selection: Challenges, Promises, and Beyond  
Yuzheng Hu, Pingbang Hu, Han Zhao, Jiaqi W. Ma
The 38th Annual Conference on Neural Information Processing Systems (NeurIPS 2024)
- [P11] Localize-and-Stitch: Efficient Model Merging via Sparse Task Arithmetic  
Yifei He, **Yuzheng Hu**, Yong Lin, Tong Zhang, Han Zhao
Transactions on Machine Learning Research (TMLR 2024)
- [P12] SoK: Privacy-Preserving Data Synthesis  
Yuzheng Hu*, Fan Wu*, Qinbin Li, Yunhui Long, Gonzalo Munilla Garrido, Chang Ge, Bolin Ding, David Forsyth, Bo Li, Dawn Song
The 45th IEEE Symposium on Security and Privacy (S&P 2024)
- [P13] HPL-ViT: A Unified Perception Framework for Heterogeneous Parallel LiDARs in V2V 
Yuhang Liu, Boyi Sun, Yuke Li, **Yuzheng Hu**, Fei-Yue Wang
The 41st IEEE International Conference on Robotics and Automation (ICRA 2024)
- [P14] Revisiting Scalarization in Multi-Task Learning: A Theoretical Perspective 
Yuzheng Hu, Ruicheng Xian, Qilong Wu, Qiuling Fan, Lang Yin, Han Zhao
The 37th Annual Conference on Neural Information Processing Systems (NeurIPS 2023)
- [P15] Understanding the Impact of Adversarial Robustness on Accuracy Disparity  
Yuzheng Hu, Fan Wu, Hongyang Zhang, Han Zhao
The 40th International Conference on Machine Learning (ICML 2023)
- [P16] Actor-critic is implicitly biased towards high entropy optimal policies 
Yuzheng Hu*, Ziwei Ji*, Matus Telgarsky*
The 10th International Conference on Learning Representations (ICLR 2022)
- [P17] Towards Understanding the Data Dependency of Mixup-style Training  
Muthu Chidambaram, Xiang Wang, **Yuzheng Hu**, Chenwei Wu, Rong Ge
*The 10th International Conference on Learning Representations (ICLR 2022, **Spotlight**)*
- [P18] Don't Waste Your Bits! Squeeze Activations and Gradients for Deep Neural Networks via TinyScript 
Fangcheng Fu, **Yuzheng Hu**, Yihan He, Jiawei Jiang, Yingxia Shao, Ce Zhang, Bin Cui
The 37th International Conference on Machine Learning (ICML 2020)

- MANUSCRIPTS
- [M1] A Unified Theory of Random Projection for Influence Functions  
Pingbang Hu*, **Yuzheng Hu***, Jiaqi W. Ma*, Han Zhao*
 - [M2] A Survey of Data Attribution: Methods, Applications, and Evaluation in the Era of Generative AI 
Junwei Deng*, **Yuzheng Hu***, Pingbang Hu*, Ting-Wei Li*, Shixuan Liu*, Jiachen T. Wang, 21 others, Jiaqi W. Ma

INVITED TALKS	Differentially Private Conditional Text Generation with RL-Boosted Control	
	- Google Athena seminar - Google ML Privacy seminar (Host: Natalia Ponomareva)	Sep. 2025 Sep. 2025
	Empirical Privacy Variance	
	Google ML Privacy seminar (Host: Matthew Jagielski)	Mar. 2025
EMPLOYMENT	Google Research <i>Research Scientist</i>	New York, NY, US May. 2026 – now
INTERNSHIP EXPERIENCE	Google Research <i>Research Intern, Hosts: Zheng Xu, Ryan McKenna, Da Yu; Manager: Peter Kairouz</i>	Mountain View, CA, US May. 2025 – Nov. 2025
	Developed a metadata-conditioned framework for DP synthetic text generation with [P2] and without [P1] fine-tuning	
	Jane Street Group, LLC <i>Quantitative Trading Intern</i>	New York, NY, US May. 2024 – July. 2024
	- Ranked 2/200 (1/150 among interns) in the Tenth Annual Figgie World Championships - Participated in mock trading games and worked on two desk projects (options and equities)	
	Damo Academy, Alibaba Group (U.S.) Inc. <i>Research Intern, Mentor: Bolin Ding</i>	Bellevue, WA, US May. 2022 – Aug. 2022
	- Worked on the intersection of privacy and law - Wrote a systematization of knowledge paper on privacy-preserving data synthesis [P12]	
RESEARCH EXPERIENCE	Simons Institute for the Theory of Computing <i>Visiting Graduate Student, Program: Modern Paradigms in Generalization </i>	Berkeley, CA, US Aug. 2024 – Dec. 2024
	Quantified the discrepancy between differential privacy and memorization in LLMs [P8], with Pritish Kamath and Chiyuan Zhang	
	University of Illinois at Urbana-Champaign <i>Graduate Research Assistant, Advisor: Han Zhao</i>	Urbana, IL, US Feb. 2022 – now
	- Thesis research: data attribution [P5] [P6] [P10] [M1] [M2], with Jiaqi Ma - Other works: model merging [P7] [P11]; representational limitations of scalarization in multi-task learning [P14]; trade-off between robustness and fairness [P15]	
	University of Illinois at Urbana-Champaign <i>Graduate Research Assistant, Advisor: Matus Telgarsky</i>	Urbana, IL, US Aug. 2021 – Dec. 2021
	Proved that actor-critic converges to high entropy optimal policy without explicit regularization [P16]	
	Duke University <i>Undergraduate Research Assistant, Advisor: Rong Ge</i>	Remote Jun. 2020 – Feb. 2021
	Demonstrated the data-dependency of mixup [P17]	

School of Intelligence Science and Technology, Peking University <i>Undergraduate Research Assistant, Advisor: Liwei Wang</i>		Beijing, China Aug. 2019 – Jun. 2020
Identified two non-local properties of the loss landscape of neural networks		
School of Computer Science, Peking University <i>Undergraduate Research Assistant, Advisor: Bin Cui</i>		Beijing, China Aug. 2018 – Aug. 2019
Introduced a distribution-aware, non-uniform quantization algorithm for multilayer perceptrons [P18]		
TEACHING EXPERIENCE	UIUC CS 591 – Machine Learning Reading Group	Spring 2024, Fall 2024
	UIUC CS 598 – Transfer Learning	Spring 2023
	UIUC CS 498 – Trustworthy Machine Learning	Fall 2022
SERVICES	Seminar Co-Organizer: UIUC Machine Learning Seminar 2024 – 2026 Workshop Organizer and Primary Contact: DATA-FM @ ICLR 2026 Area Chair: NeurIPS 2026 Position Paper Track Program Committee: TPDP (2026, 2025) , ICLR SynthData (2025) , ACL L2M2 (2025) Reviewer: NeurIPS (2022-2025), ICML (2023-2026), ICLR (2024-2026), AISTATS (2024), TMLR	
HONORS AND AWARDS	Top/Outstanding Reviewer: ICML 2026 (Top 25%), ICML 2025 (Top 2%), NeurIPS 2025 (Top 8%), NeurIPS 2023 (Top 10%)	
	ICML 2026 Oral (Top 0.7%, 168/23,918)	2026
	NeurIPS 2025 Oral (Top 0.3%, 77/21,575)	2025
	Saburo Muroga Endowed Fellowship (Top 4% in UIUC)	2021-2022
	Exceptional Award for Academic Innovation (Top 1% in Peking University)	2020
	Elite Undergraduate Training Program of Applied Math (Top 15% in SMS)	2019-2021
	Invited Participant, China TST for the 58th International Mathematical Olympiad (IMO)	2017
	Gold Medal, 32nd Chinese Mathematical Olympiad (CMO) (Top 60 in China)	2016
TRAVEL GRANTS	TPDP Student Travel Funding (\$825)	2025
	IEEE S&P Student Travel Grants Award (\$1,000)	2024
	NeurIPS 2023 Scholar Award (\$2,000)	2023
SKILLS	Programming Frameworks & Packages	Python, Shell Script, $\text{\LaTeX}$, MATLAB PyTorch, HuggingFace